A. Experimental Setup

This appendix provides the detailed human-evaluation protocol supporting the concise results reported in Section 4.6. The main paper reports the compact Textual Alignment and Image Alignment summary, while the full questionnaire, Likert scores, and preference counts are placed after the references. This separation keeps the main paper focused on the central claims while preserving the information needed to reproduce the user-study aggregation.

A.1. Human Evaluation Protocol

We conducted a pilot human evaluation with ten participants (P1-P10). Participants compared five anonymized visual story methods labeled Method A-E. Method A corresponds to DCEE-CausalVerse, while the remaining methods correspond to baseline-generated sequences. To reduce method-name bias, the survey used only anonymized method labels. Each participant was shown a reference image and a textual appearance description of the target white-bear protagonist, the six frame-level story captions, and the generated six-frame sequences for all methods. Participants rated each method using eight five-point Likert questions, where 1 indicates very poor and 5 indicates excellent. They also selected the best and second-best overall visual story. We compute the final preference score as 2 x First Choice + 1 x Second Choice.

For compact reporting in the main paper, we aggregate the eight Likert questions into two ViSTA-style criteria. Textual Alignment averages Q3-Q6: caption reflection, object/action visibility, ending-emotion delivery, and emotion-cause explainability. Image Alignment averages Q2 and Q8: reference-image adherence and overall visual quality/completeness. We do not include a separate Consistency column in the main text, because the current paper emphasizes whether the visual sequence reflects the causal event-evidence structure and image-level visual quality; the detailed consistency-related questions are still reported below.

A.2. Questionnaire

Table A1. Human evaluation questionnaire and evaluation dimensions.

Question	Dimension	Question used for evaluation	Relevance to DCEE-CausalVerse
Q1	Character consistency	Does the protagonist look like the same character across Frames 1-6?	Identity continuity
Q2	Reference adherence	Does the generated protagonist follow the reference image appearance?	Reference-based character fidelity
Q3	Caption reflection	Does each frame reflect its corresponding caption?	Frame-level story grounding
Q4	Object/action visibility	Are key objects and actions such as the honey jar, branches, searching, hiding, discovering, and eating visible?	Event and evidence visibility
Q5	Ending emotion delivery	Is the happy/joyful emotion clearly felt in the latter part of the sequence?	Emotional progression
Q6	Emotion-cause explainability	Can the viewer understand why the protagonist feels happy at the end based only on the generated images?	Central event-evidence causal objective
Q7	Style consistency	Is the visual style consistent across frames?	Cross-frame visual coherence
Q8	Overall visual quality	Are the generated frames high quality in terms of background, completeness, and overall finish?	Visual appeal and completeness

Table A1 shows that the questionnaire directly evaluates the main claims of DCEE-CausalVerse. Q3 and Q4 measure frame-level story grounding and visible event/evidence realization. Q5 and Q6 measure whether the target happy ending is delivered and whether the viewer can infer its cause from the image sequence. Q6 is therefore the closest human-evaluation counterpart of the proposed event-evidence-grounded emotional storytelling objective.

A.3. Detailed Human Evaluation Results

Table A2. Detailed Likert results averaged over ten participants. Method A denotes DCEE-CausalVerse.

Method	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Overall	Rank
Method A (Ours)	2.90	3.10	4.50	4.70	4.60	4.60	2.20	4.60	3.90	1
Method B	3.20	3.10	1.60	3.40	4.00	2.50	3.30	3.40	3.06	3
Method C	3.80	3.60	2.90	4.20	4.00	4.10	3.60	4.20	3.80	2
Method D	4.70	3.00	2.40	2.30	2.40	2.20	4.30	2.10	2.93	4
Method E	4.70	3.00	1.90	2.00	2.40	1.90	4.30	2.10	2.79	5

Table A2 reports the full Likert results. Method A obtains the highest overall mean score (3.90), and is strongest on the criteria most closely tied to our research objective: caption reflection (Q3 = 4.50), object/action visibility (Q4 = 4.70), ending-emotion delivery (Q5 = 4.60), emotion-cause explainability (Q6 = 4.60), and overall visual quality (Q8 = 4.60). Although Method A is not the strongest on every identity/style question, it is judged strongest on the story-grounding and causal-emotion dimensions that define ending-controllable causal visual storytelling.

Table A3. Aggregated ViSTA-style human-evaluation criteria used in the main paper.

Model	Textual Align. ↑	Image Align. ↑	Definition
Method A (Ours)	4.60	3.85	Textual: avg(Q3-Q6); Image: avg(Q2,Q8)
Method B	2.88	3.25	Textual: avg(Q3-Q6); Image: avg(Q2,Q8)
Method C	3.80	3.90	Textual: avg(Q3-Q6); Image: avg(Q2,Q8)
Method D	2.33	2.55	Textual: avg(Q3-Q6); Image: avg(Q2,Q8)
Method E	2.05	2.55	Textual: avg(Q3-Q6); Image: avg(Q2,Q8)

Table A3 defines the two main-paper aggregates. Method A ranks first on Textual Alignment, while Method C is slightly higher on Image Alignment.

Table A4. Final preference results. Preference points are computed as 2 x First Choice + 1 x Second Choice.

Method	First choice	Second choice	Preference points	Preference rank
Method A (Ours)	6	3	15	1
Method B	0	1	1	3
Method C	4	6	14	2
Method D	0	0	0	4
Method E	0	0	0	4

Table A4 reports the final preference results. Method A receives six first-choice votes and three second-choice votes, yielding the highest preference score of 15 points. Method C is competitive with 14 points, but Method A is selected as the best visual story by the majority of participants. These results reinforce the Likert analysis: DCEE-CausalVerse is perceived as the strongest method overall, particularly because viewers can follow the captioned events, observe the relevant objects/actions, and understand why the protagonist reaches the final happy emotion.

B. Additional Method Details

This appendix preserves method details that are useful for reproducibility but too long for the main body. The main paper reports the conceptual method comparison, while this appendix keeps the compact pipeline overview and complete inference algorithm so that the planning, control construction, generation, reranking, retry, and memory-update stages can be traced step by step.

Table B1. Compact step-by-step overview of the DCEE-CausalVerse pipeline.

Stage / role in the pipeline
Input specification — receives the initial image/text prompt, protagonist description, and target ending emotion.
Global Story Bible — stores story-wide identity, world, object, style, and affective constraints.
DCEE-Tree planning — explores multiple Desire-Conflict-Event Chain-Ending Emotion routes.
Route selection — selects the route with the strongest causal and emotional support.
Canonicalization — converts abstract events into drawable subjects, actions, evidence, and world states.
Causal memory and anchors — retrieves event-emotion memory and preserves character, evidence-object, and world anchors.
Butterfly controller — expands visual hypotheses, fuses them with memory/anchors, and decodes factorized controls.
Factor branches — separates Character, World, Emotion, Event, and Evidence controls.
Cross-attention injection — converts factor branches into SDXL cross-attention control tokens.
Generation and reranking — generates candidates autoregressively and selects frames by event/evidence/emotion and consistency scores.
Final assembly — outputs the visual story sequence, storyboard, control packets, and evaluation logs.

Algorithm 1. DCEE-CausalVerse inference procedure.

Input: x: input premise; a: protagonist description; e*: target ending emotion; K: number of frames; N: number of image candidates per frame
Output: final image sequence Y1:K, storyboard S1:K, selected DCEE plan P*, visual control packets, and evaluation logs
B ← **BuildStoryBible**(x, a, e*)
R ← **GenerateDCEERoutes**(B) // multiple Desire-Conflict-Event routes
for each route rj in R **do**
    Ej ← **ExpandEventChain**(rj)
    vj ← **EvaluateRoute**(rj, Ej, e*) using causal coherence, emotion alignment, visual evidentiality, diversity, and narrative coherence
end for
*P** ← arg maxj vj
S1:K ← **Canonicalize**(P*) into drawable subjects, actions, evidence objects, world states, and emotion cues
M0, A0 ← **Initialize** causal sink memory and DCEE anchor bank
for k ← 1 to K **do**
    qk ← event-emotion query from current DCEE frame fk and storyboard Sk
    Mk ← **RetrieveCausalMemory**(qk, Mk−1) using event, conflict, clue, emotion-cause, identity, and world relevance
    Zk ← **Decoder1**(fk) = {zk1, zk2, ..., zkN} // narrative expansion into visual hypotheses
    hk ← **CausalFusionEncoder**(Zk, Mk, Ak, Ck)
    Bk ← **Decoder2**(hk) = {bchar, bworld, bemotion, bevent, bevidence}
    Haug ← [Hprompt; Hchar; Hworld; Hemotion; Hevent; Hevidence]
    {yk,n}n=1..N ← **SDXLGenerate**(Haug)
    *y*k* ← **Rerank**({yk,n}) using event grounding, evidence visibility, emotion visibility, event-emotion causal consistency, identity consistency, world consistency, and image quality
    if y*k fails event/evidence/emotion thresholds **then**
        Strengthen the weak control branch and regenerate candidates for frame k
    end if
    Mk+1, Ak+1 ← **UpdateMemoryAndAnchors**(y*k, Sk, Bk)
end for
Return Y1:K = {y*1, ..., y*K}, S1:K, P*, control packets, and evaluation logs

Algorithm 1 gives the complete inference procedure, from DCEE route generation and selection to autoregressive frame generation, reranking, retry, and memory update.

C. Additional Experimental Details

This appendix lists the baseline and metric definitions used in the experiments. The main paper reports a compact experimental summary, while the appendix clarifies what each comparison method tests and how each automatic or diagnostic metric relates to event-grounded causal visual storytelling.

Table C1. Baselines used to evaluate DCEE-CausalVerse.

Baseline / comparison purpose
Direct LLM + SDXL — tests whether direct prompt engineering can support ending-controllable visual storytelling.
SDXL Prompt-only — measures local caption-to-image rendering without sequence-level memory or structured controls.
LLM Storyboard + SDXL — isolates the value of DCEE planning over generic LLM storyboarding.
ViSTA-style history — compares causal sink memory with relevance-based history conditioning.
ConsiStory / StoryDiffusion-style backend — tests whether identity consistency alone is sufficient for causally justified ending control.
DCEE w/o Evidence Branch — quantifies the contribution of explicitly modeling visible causal evidence.
DCEE w/o Tree Planning — measures the importance of multi-route DCEE-Tree exploration.
Full DCEE-CausalVerse — uses all proposed modules: planning, canonicalization, memory, anchors, Butterfly control, reranking, and retry.

Table C2. Automatic and diagnostic metrics used in the evaluation protocol.

Metric / use in this paper
CLIP-T ↑ — measures frame-caption semantic alignment.
TIFA ↑ — checks whether required objects, attributes, and relations are visible using VQA-style faithfulness.
CLIP-I ↑ — measures visual similarity and consistency across selected story frames.
Ending Emotion Accuracy ↑ — evaluates whether the final frame expresses the requested ending emotion.
Event Grounding Score ↑ — diagnoses whether the planned event/action is visually depicted.
Evidence Visibility Score ↑ — checks whether the causal clue or evidence object is visible.
Event-Emotion Causal Score ↑ — measures whether the visible event plausibly explains the protagonist emotion.
Counterfactual Trajectory Diversity ↑ — tests whether different target emotions produce distinct event chains from the same premise.

D. Additional Experimental Results

To provide a more comprehensive evaluation of the proposed framework, this appendix reports case-level automatic results and qualitative comparisons for five additional input configurations. Each configuration combines a reference image, a target ending emotion, and a genre. Within each configuration, DCEE-CausalVerse and the four baselines are evaluated under the same six-frame story specification. These additional experiments are intended to stress-test whether the proposed causal planning and event-evidence controls remain useful across different protagonists, affective targets, and story genres.

Unlike the comparison methods, which primarily perform direct text-to-image generation from predefined frame-level descriptions, DCEE-CausalVerse addresses the broader text-to-story-to-image task by first constructing a causally structured narrative and then generating the corresponding image sequence. Despite this additional challenge, our method maintains competitive automatic performance and demonstrates strong generation quality in the human evaluation, achieving the highest overall rating and preference score, particularly for story alignment, event visibility, emotion delivery, and emotion-cause explainability. These results show that DCEE-CausalVerse enables a more creative and expressive generation process without substantially sacrificing image quality, highlighting its strong performance even under the more challenging end-to-end setting.

D.1. Case-Level Automatic Results

Tables D1-D5 report the automatic metrics for five input configurations: owl + anger + mystery, cat + happy + comedy, dog + fear + thriller, boy + happy + coming-of-age drama, and girl + sad + slice-of-life drama. These case-level experiments supplement the overall evaluation of the study by examining performance across different protagonists, target ending emotions, and genres.

Table D1. Automatic results for Input Combination (owl + anger + mystery).

Method	CLIP-T (Mean) ↑	CLIP-I (Mean) ↑	TIFA (Mean) ↑
Ours	0.3174	0.7188	0.5000
SDXL	**0.3308**	0.7093	**0.6667**
IP-Adapter	0.3184	**0.7960**	0.5667
StoryDiffusion (Emotion)	0.3198	0.7557	0.4000
StoryDiffusion (no Emotion)	0.3048	0.7560	0.4333

Table D2. Automatic results for Input Combination (cat + happy + comedy).

Method	CLIP-T (Mean) ↑	CLIP-I (Mean) ↑	TIFA (Mean) ↑
Ours	0.2899	0.7326	0.5667
SDXL	**0.3177**	0.7259	0.6000
IP-Adapter	0.3104	**0.7938**	**0.6667**
StoryDiffusion (Emotion)	0.3167	0.7462	0.6333
StoryDiffusion (no Emotion)	0.2934	0.7528	0.6000

Table D3. Automatic results for Input Combination (dog + fear + thriller).

Method	CLIP-T (Mean) ↑	CLIP-I (Mean) ↑	TIFA (Mean) ↑
Ours	0.2601	0.6778	0.5667
SDXL	**0.2853**	0.6309	**0.6667**
IP-Adapter	0.2833	**0.7478**	0.5333
StoryDiffusion (Emotion)	0.2768	0.7275	0.4000
StoryDiffusion (no Emotion)	0.2405	0.7351	0.4000

Table D4. Automatic results for Input Combination (boy + happy + coming-of-age drama).

Method	CLIP-T (Mean) ↑	CLIP-I (Mean) ↑	TIFA (Mean) ↑
Ours	0.2805	0.6176	0.5000
SDXL	0.2863	0.6508	0.7333
IP-Adapter	0.2981	**0.7449**	**0.8333**
StoryDiffusion (Emotion)	**0.3006**	0.7285	0.7000
StoryDiffusion (no Emotion)	0.2810	0.7315	0.6667

Table D5. Automatic results for Input Combination (girl + sad + slice-of-life drama).

Method	CLIP-T (Mean) ↑	CLIP-I (Mean) ↑	TIFA (Mean) ↑
Ours	0.2878	0.6636	0.5667
SDXL	0.2927	0.6640	0.6000
IP-Adapter	0.2996	**0.7642**	0.6333
StoryDiffusion (Emotion)	0.3138	0.7010	**0.7000**
StoryDiffusion (no Emotion)	**0.3184**	0.7096	**0.7000**

Higher values are better; bold values indicate the best result in each metric column. Ties are bolded jointly.

D.2. Qualitative Comparisons Across Five Input Configurations

Figures D1-D5 show the generated six-frame sequences for DCEE-CausalVerse, SDXL Prompt-only, IP-Adapter, StoryDiffusion with explicit emotion conditioning, and StoryDiffusion without emotion conditioning. Rows denote methods and columns denote Frames 1-6. The reference image, target ending emotion, and genre are fixed within each input configuration.

Figure D1. Qualitative comparison for Input Configuration (owl reference image; anger target ending; mystery genre).
Generated story abstract :
“A brown owl studies a star map on a moonlit clock tower ledge. The brown owl looks through a brass telescope. The brown owl proudly marks a bright comet. The brown owl hears a sudden wind knock the telescope over. The brown owl sees the star map tear. The brown owl stomps on the clock tower ledge with fluffed feathers and an angry glare.”

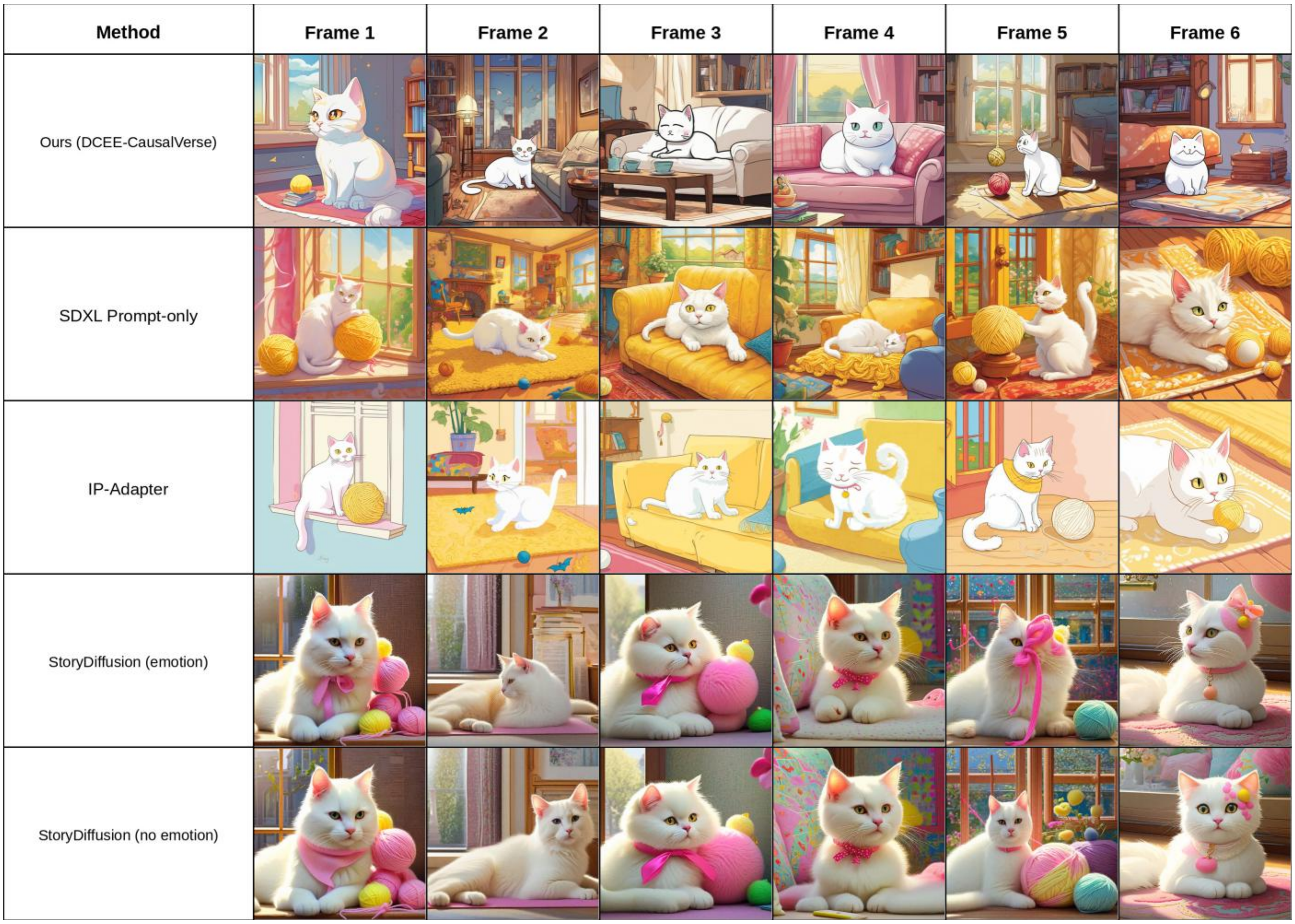

Figure D2. Qualitative comparison for Input Configuration (cat reference image; happy target ending; comedy genre).
Generated story abstract :
"A white cat with a pink collar finds a yellow yarn ball beside a sunny window. The white cat bats it across the living room. The white cat looks startled when it rolls under a sofa. The white cat squeezes under the sofa. The white cat pulls the yarn ball free. The white cat happily tumbles onto a soft rug to play in warm sunlight."

Figure D3. Qualitative comparison for Input Configuration (dog reference image; fear target ending; thriller genre).
Generated story abstract :
"A puppy happily plays with a red ball in a quiet yard until the ball bounces away and stops before a dark garage. The puppy approaches the garage. The puppy picks up the red ball. The puppy freezes at the long shadow. The puppy turns away. The puppy runs back to the yard trembling in fear."

Figure D4. Qualitative comparison for Input Configuration (boy reference image; happy target ending; coming-of-age drama genre).
Generated story abstract :
“A boy walks into a sunny park holding a red kite. The boy tries to fly it near a tall tree. The boy struggles when the kite string tangles in the wind. The boy gently frees the kite from a low branch. The boy ends in happiness while watching it soar above the park. The boy reaches the ending emotion: happy”

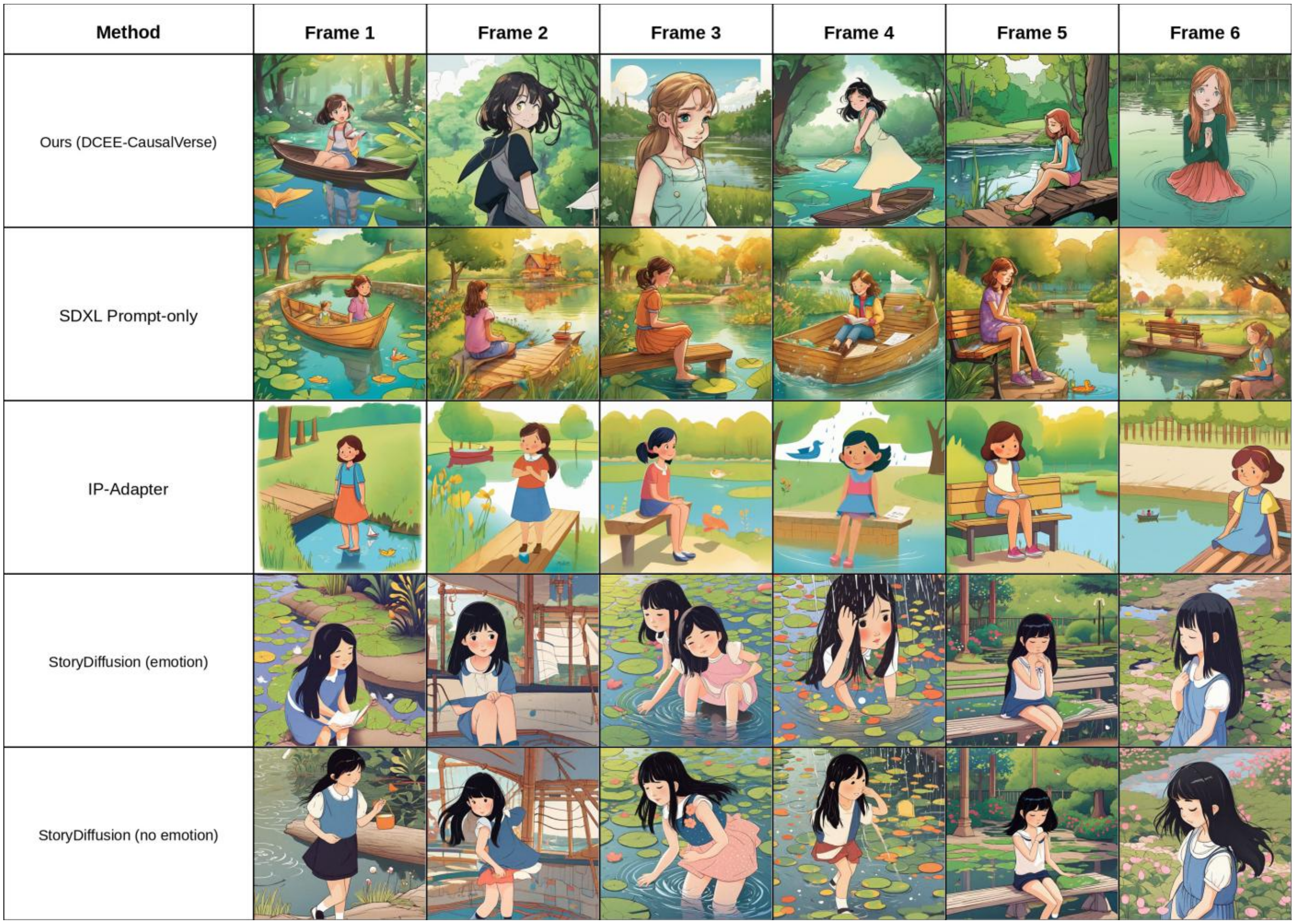

Figure D5. Qualitative comparison for Input Configuration (girl reference image; sad target ending; slice-of-life drama genre).

Generated story abstract :

“A girl carries a small paper boat to a quiet pond. The girl hopes to watch it sail. The girl places it gently on the water. The girl sees the paper boat soften and sink in the rain. The girl ends in sadness while sitting alone on a park bench beside the pond. The girl reaches the ending emotion: sad”